

Investigating the Gender Gap and the Effect of Choice Overload on Charitable Behaviour

A PSY916 Project
Presented to the Graduate School
of
for the degree of
Behavioural and Economic Science
University of Warwick

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31 August 2022

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Target Journal -Journal of Consumer Psychology

Abstract

The extant literature on charitable giving postulates that a gender difference exists in donation and donation amount among men and women. Women have usually been found to give at significantly higher levels than men because of several factors explained by multiple theories from psychology, economics and sociology. The current study investigates the gender gap in giving behaviour that has already been established in the extant literature using different donation tasks that differ in the type of charities. Additionally, the study also investigates how the assortment size of the contribution options influences people's giving behaviour using three different choice sets that with 4 options , 12 options and 32 options . The study also investigates whether information processing systems influence giving behaviour or not using the CRT. Overall, there was a gender difference in giving behaviour with women donating at slightly higher levels than men with age and income being the most consistent moderators. Also, a reversal of choice overload was found with donors donating more in the large choice set as opposed to the small and reporting higher levels of choice satisfaction in large choice sets. Lastly, the study did not find a significant relationship between information processing system and giving behaviour.

Keywords : Giving behaviour, Gender-gap, Choice Overload, CRT

1 Introduction

The tendency for women to give more than men to charitable organization's is the most robust finding in research on charitable giving across multiple cultural backgrounds. This claim is further supported by evidence from generalizable individual studies which show that when demographic variables (e.g.- wealth, education, marital status) are controlled for, single women donate more to charity than similarly situated men(Mesch et al.,2015). Not only that, but women are also more likely than men to donate to every charitable subsector with the only exception being giving to athletic teams and recreation activities whereby men are more likely than women to give to these subsectors. Also, women support higher number of causes than men by spreading their giving across more organization, whereas men tend to focus on a small range of organizations. It has also been found out that women are twice more likely than men to volunteer and volunteer more hours for charitable organizations. Therefore, the general consensus across the charitable giving literature is that women are more likely to give and volunteer to charities across different subsectors than men and there is a gender difference in giving behaviour.

This gender differences in giving arises due to four important factors that can be explained by several theories from multiple disciplines ranging from psychology to economics and sociology (Mesch et al.,2015). Firstly, research from psychology suggests that women give more than men because they are more empathetic. Empathy is a trait which is linked to altruism and helping behaviours towards the needy. In addition to empathy, it has been found that men's giving behaviour is motivated by the benefits from being charitable. Specifically, men are more likely to give if they receive a tax break from the donation, whereas women will give even if there is no tax break from the donation. This is known as the Price of Giving theory (Andreoni and Vesturlund ,2001) which postulates that men and women are equally altruistic, but men are more altruistic when they receive a personal benefit from giving while women's altruism is not determined by their personal benefit from giving. Therefore, research from psychology concludes that women give to charity more than men because they are more altruistic. The only exception to this finding are the situations whereby men receive a personal benefit from donation and are more generous than women. Secondly, research from sociology suggests that gender differences in giving are also influenced by social norms, social distance, and social capital. One reason why women and men contribute differently to charity is because men consider social norms into giving while women do not. Specifically, men are more generous if their giving behaviour aligns with the average group behaviour, while

women's giving behaviour is not moderated by group behaviour. (Croson and Gneezy, 2009; Meier, 2007). Instead, women's giving behaviour is related to social distance. Social distance refers to the social separation between the donor and receiver (Cox and Deck, 2006). When women feel they are socially close to the receiver, they donate more and when they feel socially distant from the receiver, they give less. Another sociological factor which impacts giving behaviour in men and women is social capital (Wang and Graddy, 2008). Social capital refers to an individual's physical social network and trust for others. Previous research findings show that social capital is highly correlated with giving behaviour in women rather than men. Brown and Farris (2007) found that social capital influences and leads to more helping behaviour and giving to charitable organizations in women and then men. The research from sociology concludes that gender differences in giving behaviour arises because men's giving behaviour is influenced by social norms whereas women's giving behaviour is determined by social distance and their social capital.

Thirdly, research from economics suggests that the gender differences in charitable behaviour arises because of differences in cooperative behaviour, risky preferences, and inequality aversion. Eckel and Grossman (2001) found that women are more likely to cooperate with other players in an ultimatum game than men and infer that women are more generous than men because of their desire to cooperate. In terms of risk preferences, women tend to be more risk averse than men and view risky situations as threats rather than challenges. This can explain why women are more egalitarian in their giving than men, it's because they view donating to a single organization as a riskier choice and perceive the situation to be threatening. Therefore, research from economics concludes that women donate more to charities because they have a stronger desire to cooperate than men and spread their donation more evenly across multiple organizations than men because they are more risk averse than men.

Lastly, there are many demographic factors like income, wealth, marriage, and religious affiliations which lead to more complex gender differences in giving among men and women. List (2004) postulates that the donors age has varying impact by gender. Women tend to be consistent in their giving to charities over time across age while on the other hand men tend to give less when they are young and increase their giving behaviour as they are older. In other words, men's giving significantly increases with age while women's giving doesn't significantly increase with age. Marriage also has varying impact by gender and overall marriage is positively associated with giving as couples donate more than single men or women. Also, men tend to increase their giving after marriage because it socialises them to be more philanthropic. Religious affiliations also impact giving in men and women. Hughes and

Lukesetic (2008) found that married couples donate less across different charities if they are not religious. Additionally, individual studies find that religious single men are more likely to donate than non-religious men (Masch et al.,2010). Religion doesn't play an important role in giving among single women as they are more likely to give anyways because they are more altruistic. Therefore, it can be concluded that religion has a positive impact on giving among single men and married couples. However, the findings on religion and giving behaviour are not conclusive because these studies focus on a small donor population along with focusing on a very few religions, mainly Christianity (Women's Philanthropy Institute [WPI], 2014). Giving behaviour also differs by income and it has been found that wealthy women are more likely to plan , budget and donate to charity than similar men. Finally, education attainment also impacts giving. With increasing educational attainment, women are entering the labour force more often and research from organisational behaviour suggests that women in the organisational sector care more about giving and make stronger commitments to charities than men (Marx,2000). Therefore, there are four main demographic factors which lead to a gender difference in giving. To conclude, there are multiple factors that generate a gender differences in charitable giving , that are explained by theories from multiple disciplines and demographics.

There are also gender differences in the amount of donation made to charitable organizations. However, unlike the robust findings on giving behaviour, the research on gender differences in donation amount lacks the same consensus in their finding, with several studies showing women donate higher amount than men, others showing men donate higher than men and some studies showing no gender differences at all. Few unique studies that focus on specific donor populations find that men donate higher amount than women to athletic teams and university. Shapiro and Ridinger (2011) found that men donated more than women to college athletics performing at the elite level in the U.S. Similarly, it has been found that men are more likely to give and donate higher amounts for recreational activities than women. In studies across the U.S. and the UK it has been found that male alumni donate more to universities than female alumni because of differences in higher educational attainment (Belfield and Beney 2000). However, the results from these studies aren't conclusive because they focus on a unique population whose findings cannot be generalized to the general population. Secondly, the results of these studies might change at the present time because they use a very outdated sample and women since that time have made significant social and economic progress which are reflected in increased labor participation, increase in wealth and educational attainment. Another set of more generalizable studies that control for demographic factors like education and income show that women donate higher amounts

than men to charitable causes across multiple cultures. Chang (2005) found that women were significantly more like to donate higher donation amounts than men in Taiwan. Similar findings have been affirmed in other countries such as the U.S., UK and the Netherlands (Piper and Schnepf; 2008; Eckel and Grossman, 2007; Beekers and Schuyt; 2008). These findings are also backed by individual studies which have found that single women donate higher amounts than men. (Brown et al., 2014). Lastly, there are few studies which find no gender differences in donation amounts. Cadsby et al., (2010) in their study used the dictator game paradigm and found that men and women are similarly altruistic and found similar levels of giving and donation amount. Similarly, Kottasz (2004) affirmed the similar findings among wealthy professionals in the UK. However, these findings cannot be generalized as they use a very small and narrow sample which cannot be generalized to the mean level of donation amount in the population. To conclude, the existence of mixed findings in the literature on donation amount demonstrates that this question needs to be answered using a larger representative sample and one of the aims of the present study is to investigate the difference in donation amounts in men and women.

One important factor that has been severely understudied in the charitable giving literature unlike giving behaviour and donation amount is giving to non-profit organizations that specifically focus on women and girl issues. This area has been understudied because surveys that focus on charitable giving do not categorize the organizations that serve to women and girl issues specifically, making tracking the donation and donation amount to these organization impossible to separate out (Women's Philanthropy Institute [WPI], 2014). Research on this area is important because the presence of women in a charitable organization that specifically focuses on a female receiver population can generate gender differences in giving among men and women (Kou et al., 2013; Themudo ;2009). Kou et al., (2013) found that giving and donation amount significantly increases in male and female donors when a charitable organization increases women presence in the organization. Similarly, Themudo (2009) found that higher levels of women's participation in a charitable organization leads to more giving and volunteering among men and women. Also, further research on this question is required because only two studies in Europe and U.S have been conducted that track donations to female receiver population only, since the inception of the Women's Funding Network (WFN) in 1985, an umbrella organization of all the charities that focus on empowering female receivers and bridge the gender differences in philanthropy. Ostrander (2004) found that organizations run by and for women and girls only receive 6 percent of the total donation given to charitable organizations in the U.S. Similarly in Europe, Shah et al., (2011) found that charitable foundations that focus only on women and girls receive 4.8

percent of the donation amount given to charities in Europe. More research is required on this topic because giving trends and donation amount to female led organization for women empowerment has changed drastically since the two studies as the WFN has rapidly grown since its inception and has over 270 charities across 30 countries in 6 continents (WFN,2013). Therefore, the lack of studies on how gender differences are mediated by female led organizations for women empowerment demonstrates a need to investigate this question in more details with a larger representative sample. Therefore, another key aim of this research is to look how giving behaviour differs in men and women donors to female centric organizations, male centric organizations, and multipurpose gender-neutral charitable organizations. The present study focuses on a specific receiver population because it is impossible to identify and categorize donation trends and amount to specific charities from survey data.

Another important factor that has been understudied in the charitable giving literature is the effect of choice overload on giving behaviour. In classical choice theory it has been postulated that more choice options within a product category is always better because it reduces search cost, improves perception of quality about a product in consumers (Berger al., 2007) and improves satisfaction with the decision outcome (Deci;1975). However, in the last two decades, a stream of research on how variety influences choice likelihood and satisfaction has challenged this axiomatic assumption of classical choice theory i.e. more options are always better for consumers (Chernev et al.,2015; Iyengar and Lepper;2000). This research stream demonstrates that having choice sets with too many options generate a choice overload which leads to adverse negative behavioural and subjective consequences on consumers .The behavioural consequences of choice overload include choice deferral (consumers are less likely to make a choice), choice reversal (switching from their initial choice), preference for small choice sets and target options which are more easily justifiable (Chernev et al.,,2015). The negative consequences of choice overload on the subjective state of a consumer are lower confidence with the chosen alternative; post decision regret and lower satisfaction with the chosen alternative (Chernev et al., 2015). Large choice sets lead to choose overload because of four important factors that are categorized as objective and subjective factors. The objective factors include choice set complexity and decision task difficulty. Choice set complexity refers to the aspects of the decision task that influences the values of the choice option and there are four choices set complexity factors that influence the impact of larger choice sets on choice overload– dominant options, option attractiveness, non-alignability and feature complementarity (Chernev et al.,2015). Dominant options are options that are superior to other options in a choice set (Huber et al.,1982). When a choice set contains a dominant option, consumer are more likely to purchase from that choice set than when it

is absent. Large choice sets leads to choice overload because when the choice set increases, the number of dominant options present in it decreases and increases the likelihood of choice deferral among consumers. Alternatively, a small choice set is more likely to contain a dominant option which increases the probability of making a choice and decreases choice deferral. Therefore, large choice sets lead to choice overload because dominant options are not present which makes the choice process complicated and difficult for the consumer. Option attractiveness refers to the overall attractiveness of the choice set from which a consumer is making a choice (Chernev et al., 2009). When large choice sets are presented, a consumer evaluates that choice set to be of lower quality i.e., the choice set is composed of non-attractive options and hence leads to choice deferral. Hence, large choice sets leads to choice overload because consumer perceive it to be of lower quality. Non-alignability refers to a situation in which a feature is present in one option and absent in another option within a choice set (Markman and Medin, 1995). When a choice set is small, it is more likely to contain options with alignable features which makes it cognitively easier for the consumer to evaluate, compare and choose between them. Alternatively, when a large choice set is presented, the number of options that have non-alignable feature increases, which causes the consumer to delay making a choice. Feature complementarity refers to the extent to which the features of a product are complimenting each other in a choice set with respect to their ability to fulfill a consumer's need (Chernev, 2005). Consumers prefer choice sets with options that have non-complementary features as the cognitive effort involved in evaluating the options and making a purchase is lower. In large choice sets, the number of options with complementarily features increases which decreases the overall attractiveness of the choice set, leading to choice deferral. Secondly, decision task difficulty refers to the structural characteristics of the decision problems and the decision task factors that influence the impact of large choice set are time constraints, decision accountability and number of attributes (Chernev, 2006). Time constraints refers to the length of time taken by a consumer to evaluate the options in a choice set. Increasing the number of options increases the evaluation period which results in a cognitive challenge and the consumer engages in a non-systematic evaluation of the alternatives in the choice set. This non-systematic evaluation of the options makes the consumer feel less satisfied and confident with the chosen option, hence leading to a choice overload (Chernev et al., 2015). Alternatively in a small choice set, a consumer is likely to engage in a systematic evaluation of the options as they don't face a time constraint and cognitive challenge in evaluating the options. Decision accountability is the process of justification used by the consumer to make sense of the choice (Chernev, 2006). When a large choice set is presented, the consumer's preference for decision accountability decreases as they find it cognitively taxing to justify their choice from a large assortment and decreases

the likelihood of making choices than compared to a small assortment of options. Lastly, the number of attributes refers to the attributes used by the consumer to differentiate between the available choice options(Scheibehenne Kleber,2009). When a large choice set is present, the number of attributes on which a product is differentiated increases, which makes it cognitively taxing for the consumer to make a choice as they have to evaluate additional information, hence leading to choice overload. To conclude, larger choice sets lead to choice overload because of higher levels of decision task difficulties and choice set complexities.

The subjective factors that are implicated in choice overload are preference uncertainty and decision goals. Preference certainty refers to the extent to which a consumer has knowledge about the choice set and well-defined preferences with respect to the decision at hand(Chernev,2003b). Preference certainty enables a consumer to reduce the complexity of the decision task as the consumer knows exactly which option from the choice set is compatible with their preferences. When small choice sets are presented, the decision task at hand is less complex structurally and the consumer finds the product .However, a large choice set is structurally complex which makes it difficult for the consumer to identify their ideal preferences, which leads to preference uncertainty, resulting in a choice overload. Therefore, when consumers are presented with large choice sets, they fail to identify their articulated preferences within the choice set, making the choice process cognitively more difficult and eventually leading to choice deferral. Secondly, decision goals refer to the degree to which consumers minimize the cognitive effort involved in making a choice among options present in an assortment (Chernev,2003b). When the choice set is large, cognitive load increases because the person finds it difficult to trade-off between all the options. Decision intent and decision focus are the decision goal factors that influence the impact of large choice sets on choice overload. Decision intent refers to whether a consumer approaches the decision task with the explicit goal of making a choice or browsing between the options with no explicit goal of making a choice (Chernev,2006). Choice overload is likely to occur in a large choice set when a consumer approaches a decision task with the explicit goal of making choices as opposed to browsing goals. Decision focus refers to whether a consumer is choosing between two choice sets or choosing between options within a choice set(Chernev,2006). When a consumer focuses on choosing an option from a specific choice set , choosing from a larger choice set is more difficult as they have to evaluate and differentiate between more options which is cognitively taxing. Therefore, along with high decision task difficulty and choice complexity, higher preferences uncertainty and more cognitive effort minimizing decisions goals lead to choice overload. To conclude, the effect of choice overload has been found in a host of product categories including everyday goods such as jams (Chernev,2005), expen-

sive luxury goods like vacation packages (Goodman and Malkoc,2012), durable luxury goods like laptops (Fasolo et al., 2009) and socially important goods such as giving to charities (Mcshane and Bockenholt,2018).

In the context of giving behaviour, another stream of research on choice overload postulates that the choice overload effect is reversed when people make choices that are beneficial to others i.e., giving to charities (Polman,2012). Specifically, when people make choices for others, they experience greater satisfaction after making a choice from a large choice set rather than a small choice set. This is because people are promotion-focused when they decide for others as opposed to being prevention-focused when deciding for themselves. When people are promotion-focused they are sensitive to the absence and presence of positive outcomes and feel more satisfied with large choice sets because it precludes errors of omissions i.e., the person perceives that with more options, good items are more likely to be available. In other words, when people choose for others, they seek for more information and alternatives with non-alignable features because they perceive that large choice sets are likely to have more good items with positive outcomes as opposed to small choice sets which will have fewer items with less positive outcomes. On the contrary, prevention-focused individuals are sensitive to the absence and presence of negative outcomes and feel less satisfied with large choice sets because it produces errors of commission i.e., the person perceives that with more options, bad items are more likely to be available(Polman,2012). In other words, when people make choices for themselves, they prefer less information and alternatives with non-alignable features as they perceive that more options are likely to have more negative outcomes which makes the choice task more complicated. Therefore, there are two streams of research on the effect of choice overload on giving behaviour with one suggesting that large choice sets have detrimental effects and leads to choice deferral and less satisfaction. On the other hand, it has been postulated that large choice sets are better as people become promotion-focused and seek out for more information and alternatives when making a decision that would benefit others. From a theoretical perspective, it is very important to understand which theory is a better predictor of giving behaviour among men and women in the general population. From a practical perspective this question is important because the number of contribution options presented by charities will play a vital role in the giving behaviour of the donor. Therefore, another aim of this study is to understand which theory of choice overload is a better predictor of giving behaviour in donors.

Lastly, another important issue that has been understudied in the context of choice overload and giving behaviour is the type of information processing system used by a donor when

evaluating the contribution options. Mainly, intuitive reasoning and deliberate reasoning are the two information processing systems used by people when evaluating information (Kahneman, 2012). Intuitive reasoning is a fast, automatic, and cognitively effortless processing of information. Deliberate reasoning is a slow, deliberate and cognitively effortful processing of information. To date, only one study by Herzstein et al., (2020) has investigated the empirical linkage between giving behaviour and information processing systems and choice overload. The study found that, when people make decisions that would benefit others, they evaluate and process the information about the contribution options intuitively rather than deliberately processing each option separately. This happens because of two reasons. Firstly, making decisions that are beneficial to others carries some uncertainty because the donor has no prior knowledge about the contribution options. And, to reduce the uncertainty, donors enact prosocial giving behaviour automatically that they have learned from prior experiences of similar situations (Rand et al., 2014). Hence, when the contribution options are presented, donors process that information intuitively because they deploy automatic prosocial behaviour that are learnt from previous experiences of the similar situation. Secondly, research shows that intuitive reasoning underlies prosocial behaviours as they are influenced by emotions rather than reason (Bear and Rand., 2018). found that prosocial behaviours are driven more by emotions and not reason in a one-shot prisoner dilemma game, implicating those prosocial giving behaviours are only influenced by emotions and people are likely to enact them intuitively when presented with contribution options. Similar findings are provided by Rand et al., (2012) and Wang et al., (2014) who show that cooperation (a type of prosocial behaviour) is also driven by intuitive reasoning rather than deliberate reasoning in one shot public good games. Therefore, when donors are presented with contribution options about which they have no prior knowledge, they evaluate the options intuitively and use the prosocial behaviours that they have learnt from similar situations and social norms to reduce the uncertainty. There are also gender differences in intuitive and deliberative reasoning that impact giving behaviour in men and women. There is robust evidence in the literature which postulates that men are more likely to engage in deliberate reasoning than women and perform better on the Cognitive Reasoning Test (CRT) (Frederick, 2005). CRT is the most popular tool for measuring people's ability to override intuitive reasoning and engage in deliberate reasoning. The CRT consists of numerical puzzles in which the intuitive numerical response is incorrect, and the person needs to inhibit that intuition and engage in deliberative reasoning that involves performing a computational sequence that leads to the correct response. Since the time the CRT has been in use, it has been seen that men outperform women in the task. In the original study, Frederick (2005) found that the average score of men was 0.50 percent higher than women. This higher average score has been replicated in most of the studies that

have investigated gender differences in CRT (Toplak et al., 2014; Primi et al., 2018). Some studies have also reported gender differences without giving any average score by controlling for individual and test characteristics (Branas-Garza et al., 2019; Cueva et al., 2016). These studies show that women are 20 percent more likely to score zero than men across all the items of the CRT. This gender gap in engaging with deliberative reasoning is moderated by a negative emotional response known as mathematics anxiety which is triggered when people solve numerical problems. Mathematics anxiety disrupts the inhibition of intuition, impairs numerical processing skills and worsens performance on numerical tasks (Juanchich et al., 2020). Since CRT is numerical in nature, mathematics anxiety is the main cause of impaired performance on it (Primi et al., 2018). Furthermore, it has been seen that women tend to experience more mathematical anxiety than men, eventually leading to a gender gap in CRT performance (Devine et al., 2014; Ferguson et al., 2015). Therefore, men engage in more deliberative reasoning and outperform women on the CRT because women experience more mathematical anxiety when performing numerical problems. Therefore, it is likely that men and women are going to evaluate the contribution options provided by charities separately because they engage in different information processing systems. Women are more likely to engage in intuitive reasoning and assess the information holistically to reduce the uncertainty about the contribution option. On the other hand, men are more likely to engage in deliberative reasoning and assess each option separately as opposed to evaluating it holistically to reduce the uncertainty about the contribution options. Therefore, this study also seeks to understand whether there are gender differences in information processing systems and whether it is a factor that influences giving behaviour or not.

To conclude, our research makes a variety of contributions to the literature on giving behaviour, gender differences and choice overload. First, we seek to understand whether the previously established gender gap in giving behaviour exists. Second, the effect of different demographic factors such as income, compassion, marital status, religion and age on giving behaviour are investigated. Third, we extend the literature on giving to organizations run by women for women by investigating whether giving behaviour changes and differs among donors when giving to female-centric organizations, male-centric organizations, and multi-purpose gender-neutral organizations using a larger representative sample. Fourth, whether the number of contribution options presented to the donor influences their giving behaviour or not and whether large choice sets are beneficial or detrimental when making prosocial choices are investigated. Fifth, whether a gender gap in intuitive and deliberative reasoning exists is investigated along with investigating whether it influences giving behaviour or not. In the sections that follow, we present an overview of the study that outlines the key

assumptions of the study pertaining to the research questions postulated. This is followed by explaining the methodology of the current study along with a presentation of the results from the survey-based experiment that we conducted to test our claims. Lastly, we discuss the implications of the findings and the future directions that can be taken by readers to improve this research area.

2 Overview

The current research examines giving behaviour in the context of gender differences and choice overload. The study investigates whether gender and other demographic factors moderates giving behaviour along with investigating whether donors donate differently when presented with female-centric organizations that are led by female to advocate women empowerment. Additionally, the study also investigates whether donation amount varies as a function of the number of contribution options (many vs few) presented to the donor. Also, the study seeks to understand whether differences in higher level cognitive reasoning (deliberative and intuitive) can be a potential factor that moderates the gender differences in giving behaviour.

H1–Women donate and donate higher amounts to charities than Men because they are more empathetic.

First, one of the most consistent predictors of giving behaviour is gender and a large body of research on giving behaviour show that women are more likely to donate and give higher amounts to charity than men. Not only that, women also tend to support higher number of causes like education, welfare, poverty, emergency care and human services as seen in their tendency to spread their contributions thinly across multiple organizations rather than giving to a single organizations like men. The only exception to this finding is giving to charities focusing on sports and adult recreation whereby men tend to donate more than women. Therefore, women tend to donate more than men across charities from different sub sectors except the ones focusing on sports and recreation.

H2–Age, Marital Status, Income , Education are demographic factors that moderate giving behaviour among all donors.

Second, it has also been suggested that giving behaviour is highly moderated and patterned by demographic factors with past findings suggesting greater giving among people with higher income, more education, older, and married people. Therefore, demographic factors have a significant amount of influence on giving behaviour among donors (Willer and Owens, Working Paper).

H3-Donors donate higher amounts to gender-specific charities than gender - neutral charities.

Third, charitable giving to organizations that specifically focus on women and girls issues has been severely understudied in the literature because its impossible to identify and separate out these donation amounts from surveys that focus on charitable giving. A few studies that have been conducted on this topic show that both men and women demonstrate a significant increase in giving and donation amount when giving to charities that focus on female receiver population. Therefore, it is likely that giving behaviour and donation amount will significantly differ when giving to gender-neutral charities and gender-specific charities that focus on women and girl issues.

H4 – Donors giving behaviour and donation amount increases from the small choice set to the large choice sets because they are promotion focused.

Fourth, the number of contribution options that are presented to the donor has an impact on giving behaviour and donation amounts. A resurgence of interest in how variety affects choice likelihood and satisfaction suggest that when consumers are presented with large choice sets, it leads to choice overload and has a negative effect on the behaviour and subjective states of the consumer such as choice deferral, choice reversal, low satisfaction, and confidence. Choice overload occurs because people find it cognitively taxing to evaluate large choice sets which makes the decision task at hand complicated and difficult. Choice overload has been found in several product categories including charities. Now, in the context of charitable giving, there has been two schools of thoughts on how choice overload influences donors' evaluation of contribution options. The first theory postulates that large choice sets have a negative impact on giving behaviour and donors will contribute smaller donation amount as the number of options increases. The other theory postulates that large choice sets have a positive impact on giving behaviour and donors contribute higher amount because people become promotion-focused when choosing for others and seek for uniqueness and variety in the choice set. The research on which theory is a more robust predictor of giving behaviour

has been understudied and only one study by Herzenstein et al., (2020) shows that promotion focused theory is a better predictor of charitable behaviour and people seek for more choices when making charitable decisions. Therefore, it is likely that donors will seek for more options when making choices about charities and there will be a significant differences in the donation amounts across a small , moderate and large choice sets with donors donating higher amounts in the large choice sets than the small choice set.

H5- Men evaluate the contribution options more deliberately than women who evaluate the options intuitively as a whole.

Lastly, there are also gender differences in information processing system which can have an impact on how the contribution options are evaluated during donation. Specifically, it has been found that men engage in deliberate reasoning more than women as measured by CRT. This occurs because women experience mathematical anxiety more than men which reduces performance on CRT by impairing inhibition of intuition and numerical processing skills. Therefore, it is likely that men and women will evaluate the contribution options separately during donation. Men are more likely to deliberately assess each option separately whereas women are more likely to assess the options holistically as a whole, which can generate differences in giving behaviour among men and women. Therefore, differences in information processing system is a potential moderator of gender differences in giving behaviour.

To conclude, the current study develops five assumptions about charitable giving based on past findings in the context of charitable giving and gender differences. The next section describes the methodology employed in the study to these five assumptions

3 Method

Sample

A survey-based experiment was conducted using Qualtrics on a random sample of globally representative respondents. The respondents were recruited using Prolific which is an online research platform for sample recruitment. The respondents were invited to take part in the study by posting the study online on Prolific for two days. In total, one hundred and fifty-two respondents (Male = 72, Female= 73, non-Binary=7) aged from 18 to 49 from across the world took part in the study. Of the one hundred fifty-seven participants that took part in the study, data from twenty-two respondents (N=22) were excluded from the study as they took less than one minute or more than one hour to complete the study. Additionally, seven respondents (N=7) were excluded from the study as they did not complete the entire study. Therefore, data from twenty-nine respondents (N=29) were excluded from the from the final analysis. Overall, data from one hundred and twenty-three respondents (Male = 60, Female=60, non-Binary=3) were used for the final analysis in the study.

Design

A 2 (Gender Neutral Treatment v/s Gender Specific Treatment) x 3 (4 options v/s 12 options vs 32 options) mixed design was used in the study. The Gender-Neutral Condition and the Gendered Specific Condition were the between-subject factors in the study. The choice task i.e., 4, 12 and 32 options are the within-subject factors in the study. Each choice task consisted of charities that focus on specific causes. The choice set with 4 charity options was the small choice set and focuses on children and adult disability issues. The choice set with 12 charity options was the medium choice set and focuses on children and adult education issues. The choice set with 32 charity options was the large choice sets and focuses on children and adult health issues.

In the Gender-Neutral Treatment, respondents completed donations task whereby they were shown an image of a charity along with what it does. In this Treatment Condition, respondents were only shown gender-neutral multipurpose charities that focuses on both male and female respondents. Respondents completed 3 donation tasks separately with a small choice set, a medium choice set and a large choice set .The small choice set consisted of 4 gender-neutral charities that focused on children and adult disability issues. The medium choice was made up of 12 gender-neutral charities that focused on children and adult education issues. The large choice set was made up of 32 gender-neutral charities that focused on children and

adult health issues.

In the Gender Specific Condition similarly, respondents completed donation tasks whereby they were shown an image of a charity along with what it does. In this treatment condition, respondents were shown both gender neutral charities and gender-specific charities that specifically focused on either male respondents or female respondents. This was done to counterbalance the two between-subjects factors. Respondents completed 3 donation tasks separately with a small choice set, a medium choice set and a large choice set. The small choice set that was shown to the respondents consisted of 1 charity that catered to male respondents only, 1 charity that catered to female respondents and 2 gender-neutral charities. The charities shown in the small choice set specifically focus on children and adult disability issues. The medium choice set consisted of 2 charities that catered to male respondents, 2 charities that catered to female respondents and 8 gender-neutral charities. The charities shown in the medium choice set focused on children and adult education issues. In the large choice set respondents were shown 3 charities that catered to male respondents only, 3 charities that focused on female respondents only and 26 gender neutral charities.

Additionally, respondents also completed some additional tests that were part of the study. First, they completed a demographic questionnaire that was used to obtain information about age, gender, ethnicity, marital status, income, education, and religiosity. Second, respondents completed a numerical real effort task whereby they completed 11 (1 practice + 10 counted) numerical questions which involved adding sequences of two-digit numbers under timed condition (Oswald et al., 2015). For each correct answer, respondents were awarded 15 ECU's and a respondent could win a maximum of 150 ECU's (10 questions x 15 ECU) if they solved all the numerical questions correctly in the real-effort task. The real effort task was used to endow participants with ECU's that could be used by them in the three donation tasks across the Gender-Neutral Condition and Gendered Condition respectively. Third, respondents also completed the CRT (Federick, 2005) which is used to measure people's propensity to engage in deliberative and intuitive reasoning. The CRT was used to understand whether information processing systems (intuitive v/s deliberative) have an impact on the evaluation of charity options in the donation task. The CRT consisted of three numerical puzzles that requires respondents to inhabit their intuitive processing and engage in deliberative processing. Appendix A gives a full list of the questions provided to the respondents in the study. Lastly, after the donation task respondents completed 12 questions from the Prosocial Personal Test Penne (2002), that was used to measure prosocial behaviours such as giving to charities. The Prosocial Personal test consisted of 12 items related to em-

pathy, social responsibility, and self-reported altruism. For each item, respondents indicated on a five-point Likert scale whether the statement was never true (coded as 1), occasionally true (coded), sometimes true (coded as 3), often true (coded as 4) and always true (coded as 5). Respondents also completed a questionnaire taken from Carrol et al., (2011) which measured choice regret and satisfaction. The questionnaire consisted of one item related to the effect of choice overload and the respondents indicate on a three-point Likert scale whether there were too many options to choose from (coded as 1), they had the right amount of option (coded as 2) and whether there are too many options to choose from (coded as 3). Appendix B gives a complete list of the items taken from the Prosocial Personal Test and the Regret/Satisfaction that were shown to the respondents. These post-experiment questionnaires were used to assess the causes of gender differences in charitable giving and the impact of choice overload on charitable giving.

Procedure

The study was conducted in four steps. First, participants completed the demographic questionnaire which was used to measure the demographic factors that impact gender differences in charitable giving. Then, respondents completed the real-effort numerical task that was used to endow ECU's that could be used by the respondents to donate to different charities in the donation tasks. Respondents completed one practice numerical question and ten counted numerical questions within five minutes. Third, respondents were randomly allocated to either the Gender-Neutral Treatment or the Gendered Treatment whereby they completed three donation tasks consisting of a small choice set, a medium and a large choice set. After each choice set, respondents completed the questionnaire on choice regret/satisfaction that assessed the effect of choice overload on the subjective state of the respondent post the choice task. Lastly, respondents completed the post-experiment questionnaire which consisted of the CRT, and the Prosocial Personal Test that measured respondent's empathy, altruism, moral reasoning, and social responsibility. On average, respondents took 45 minutes to complete the study.

4 Results

Women donate at higher levels than men because they are more empathetic

Firstly, the hypothesis that women will give at higher levels to charity than men because they are more empathetic was tested. Figure 8 presents the overall donation amount of male and female respondents across both the between-subject factors (Gender Neutral Treatment v/s Gendered Treatment) and the three within-subjects factors (4 options v/s 12 options v/s 32 options). The results show that women tend to donate at slightly higher levels than men to charities. Specifically, female respondents on average donated 27 ECUs across both the treatment conditions, while male respondents donated 24 ECUs on average across both the condition. This suggests that female respondents in the study donated higher amount than male respondents to the charity. However, a significant difference in donation amount that is moderated by gender differences was not found. The data was analyzed using one way ANOVA with donation amount as the dependent variable (DV) and gender as the independent variables (IV) to examine whether gender influences charitable giving. In contrast to the main hypothesis, the ANOVA did not show a main effect of gender, $F(6,1) = .00$, $p = .94$, indicating that the difference in giving behaviour is not moderated by gender.

Additionally, whether empathy moderates the gender differences in donation amount or not was examined. Contrary to the main hypothesis, the ANOVA did not show a main effect of empathy $F(6,3) = .59$, $P = .61$. However, there were significant differences in male and female respondent's giving behaviour who score low on empathy (See Figure 2). Specifically, male respondents who score low on empathy tend to donate 51 ECUs on average while female respondents who score low on empathy tend to donate 13 ECU's overall (See 2). There are no significant differences in donation amount among male and female respondents who scored high and moderately well on empathy. Therefore, women donated at higher levels than men but these differences were not moderated by gender differences in giving behaviour. Also, empathy didn't play an important role in moderating the gender differences in giving behaviour, except among men and women who scored low on empathy, whereby men donated significantly higher numbers of ECU's than females.

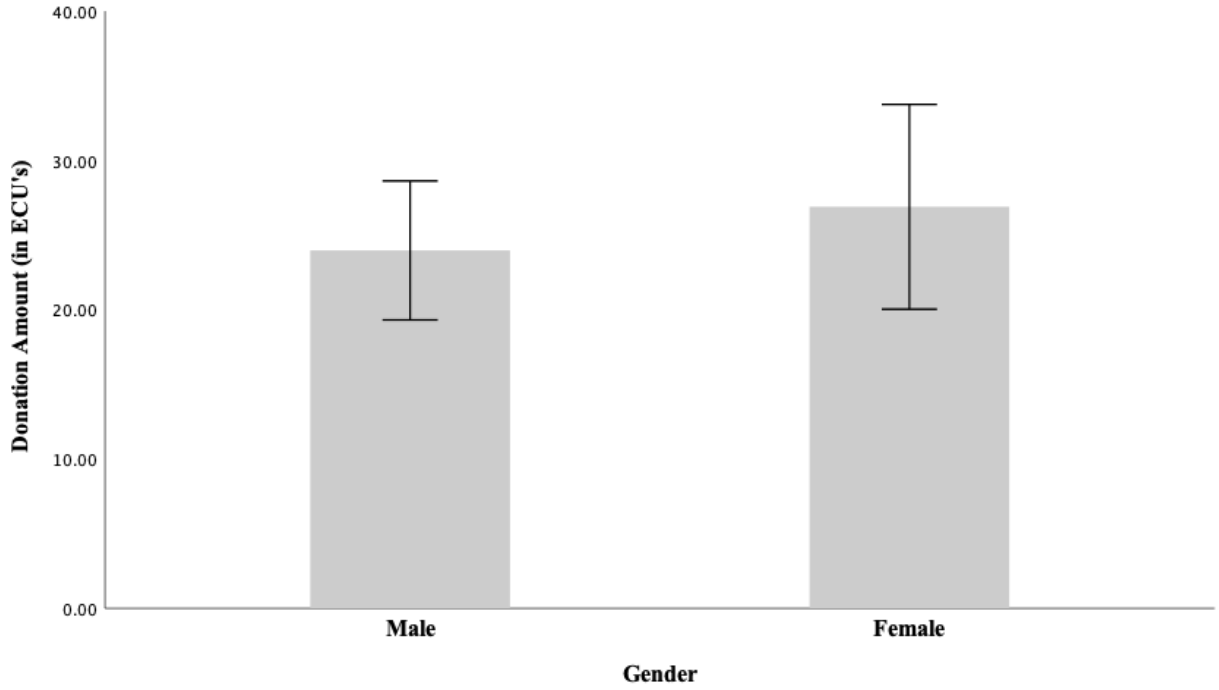


Figure 1: Overall donation by female and male respondents

Effect of demographic factors on giving behaviour

In this section the effect of demographic factors and whether it leads to differences in giving behaviour or not was examined. Firstly, donor age had a significant impact of giving behaviour. Middle-aged respondents aged from 26 to 33 ($M=29,24$) donated significantly higher amounts of ECUs across all the conditions than young ($M=24,SD=23$) and old respondents ($M=20,SD=14$). This suggests that both men and women give more when they are middle aged and give less when they are young or old. Therefore, donation among respondents from different age groups is not similar and hence donor age moderates' differences in giving behaviour. Also, female respondents ($M=27, SD=25$) in the study across all age groups donated higher ECU's than male respondents ($M=24,SD=18$) across all age groups. Secondly, we investigated whether marriage has a differential impact on charitable giving or not. Contrary to previous findings, there were no significant differences in giving among married respondents and single participants. Married respondents on average donated 27 ECU's while single respondents donated 26 ECU's on average. Therefore, in the current study, marital status did not have moderate charitable giving. Thirdly, we investigated whether income moderates giving behaviour or not. Firstly, wealthier respondents in the study significantly donated at higher levels than respondents who earned less. However, the results show that donation amount increased linearly with income. In other words, people increase their

charitable giving as income increases. Additionally, a significant interaction between income and gender differences in giving was found. Specifically, women ($M=27, SD=25$) across all age groups were significantly more likely than males to give to charity ($M=23, SD=24$). Also, wealthier women donated significantly more than similarly situated wealthy men. Therefore, income moderates giving behaviour, and has an impact on the gender gap in charitable giving. Lastly, the differential impact of education levels on giving behaviour was investigated. Contrary to prior findings, this study did not find that higher level of education leads to greater giving. Instead, in this study, respondents who completed their education to the high school level ($M=26.9, SD=27$) donated more ECU's than respondents who had completed a bachelors ($M=23.8, SD=22.5$) and masters degree ($M=26.1, SD=12.66$). Therefore, more education did not lead to higher donation levels in this sample. To conclude, in the current study the demographics factors that significantly moderated giving behaviour among respondents were income and age, while education level and marital status did not moderate giving behaviour.

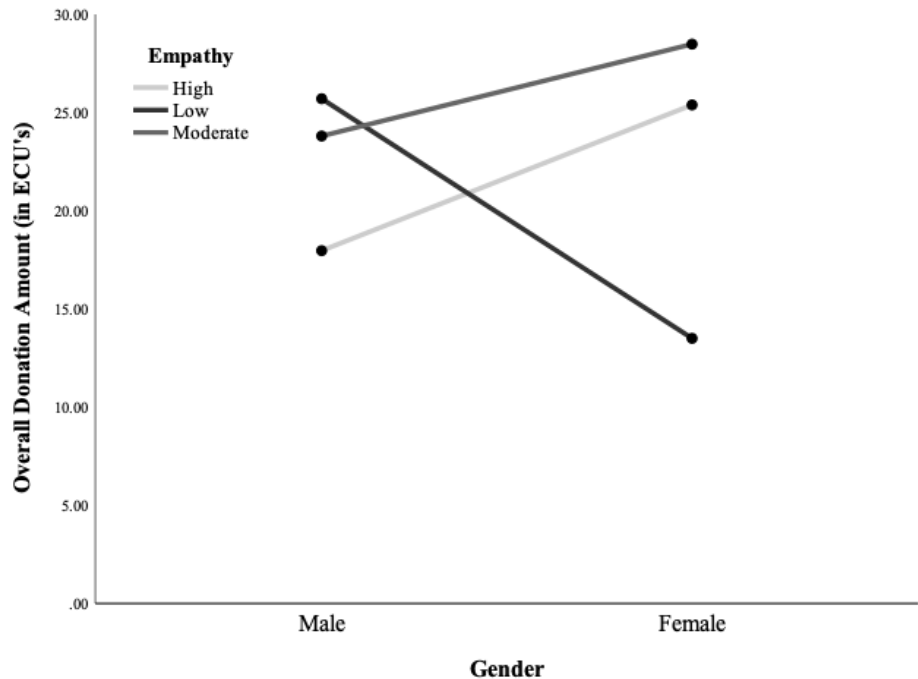


Figure 2: Mean donations levels across different empathy levels

Donors give more to gender specific charities than gender neutral charities

Table 1 presents the average donation in the Gender-Neutral Condition and the Gendered Condition across all participants. Overall, both male and female respondents donated 27.4 ECUs on average in the Gender-Neutral Condition and 24 ECUs in the Gendered Condition.

This suggests that presenting respondents with charities that specifically focus on male or female respondents did not increase their donation and instead respondents reduce their donation in the Gendered Condition.

Additionally, the gender differences in donation amount across the two conditions was also investigated. In the Gender-Neutral condition, female respondents donated 30 ECUs on average while male respondents donated 25 ECU's. In the Gendered Condition, females donated 25 ECU's and males donated 24 ECUs on average. Overall, male respondents reported a 1 percent decrease in donation in Gendered Condition, while female respondents reported a 5 percent decrease in donation as compared to the Gender-Neutral Condition. Male respondent's donation amount reduces less significantly in the Gendered Condition than female respondents. Therefore, donors donation amount and giving behaviour does not increase Gendered Condition and instead donors donate less in this condition than the Gender-Neutral Condition.

Table 1:

	<i>N</i>	<i>M</i>	<i>SD</i>
Gender Neutral Condition	46	27.4	22.9
Gendered Condition	66	24.0	22.1

Effect of assortment size on donation

Figure 3 displays respondents giving across the three-choice task with 4 options, 12 options and 32 options respectively. Overall, both male and female respondents reported a mean donation of 6 ECUs in the small choice set, 8 ECUs in the medium choice set and 13 ECUs in the large choice set. Additionally, a one-way within-subjects ANOVA was conducted to explore the impact of number of options on donation amount and giving behaviour across three choice sets (small, medium, large) (coded as Choice Overload) (See Appendix 2) . The results suggest that there was a significant effect of the number of contribution options on donation amount [$F(2,188) = 26.47, p = .001$]. Post hoc comparisons using the Bonferroni adjustments revealed that donations were significantly much higher in the large choice set ($M = 13.3, p = .001$) compared to the medium choice set ($M = 8.3, p = .001$) and the small choice set ($M = 6.2, p = .001$). Therefore, the donation amount significantly differed across the three choice sets with respondents mean donation being higher in the large choice sets as opposed to the medium and small choice set. Respondents also reported higher choice satisfaction in the large choice set ($M = 13, SD = 14$) compared to the medium choice set ($M = 9, SD = 7$) and the small choice set ($M = 6, SD = 5$). To conclude, the number of contribution options presented to the respondents had a significant effect on their charitable giving. Respondents

demonstrated higher levels of giving and choice satisfaction when making donations in the large choice set as compared to the small choice set. This postulates that there was a reversal of the choice overload effect when people make decisions that would be beneficial for others as they are promotion focused. Therefore, based on our results, the promotion focused theory seems to be a more robust predictor of the effect of variety in charitable giving.

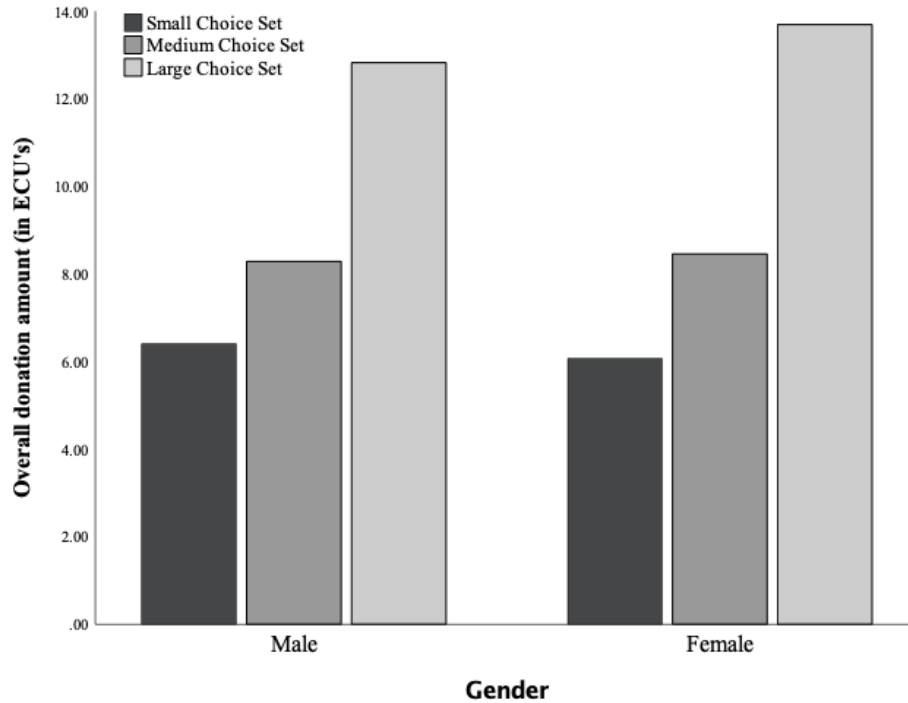


Figure 3: Donation amount across the small choice set, medium choice set and large choice set

Effect of information processing systems on donation

To test this assumption, a one-way between subjects ANOVA was conducted with donation amount as the dependent variable (DV) and gender and CRT scores as the two independent variables (IV) to assess whether information processing systems vary according to gender and whether it influences charitable giving or not. Contrary to the main effect, ANOVA did not find a main effect of CRT scores, $F(82,111)=1.05, p=.35$. Additionally, there was a non-significant CRT by gender interaction, $F(82,111)=.08, p=.92$. This suggests that respondent's ability to engage in reflective processing didn't differ by gender and did not have an influence on their charitable giving and donation amount. Figure 4 shows the statistically non-significant differences among CRT scores in relation to donation. Therefore, Hypothesis 5 can be rejected because information processing systems do not have a significant effect on charitable giving and there are no significant differences between male and female respondents in their ability to engage in reflective processing within the dataset.

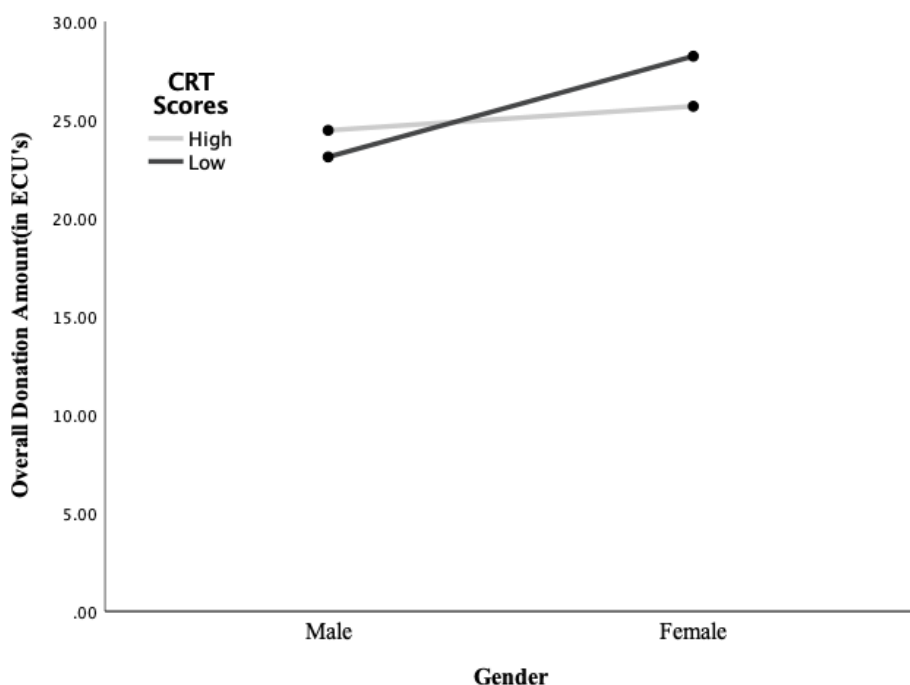


Figure 4: Mean donation amount as a function of CRT scores

5 Discussion

The results of this study show support for three out of five hypotheses that were postulated. Firstly, a gender difference in charitable giving was found with women donating at slightly higher levels than men in the donation task. This is consistent with past research on charitable giving which shows that female donate more than men across every charity sub sector. However, this gender difference in charitable giving was not mediated by different levels of empathy, which is one of the main psychological factors that contribute to this gender gap. Instead, contrary to prior findings, this study found that when men and women score low on empathy, men tend to donate at significantly higher levels than women.

Secondly, the study investigated the effect of demographic factors such as age, marital status, income, and education, which have been implicated in past research to be mediating a difference in giving behaviour. Consistent with past findings, age and income mediated the gender gap and differences in giving behaviour. Donor's age had a significant impact on donation amount, with middle aged individuals donating more than young and old individuals. This difference was probably mediated by differences in employment status. Specifically, the middle-aged individuals in the study were more likely to have a full-time job than the young and old individuals in the study. Because they had a full-time job, they were earning more than the young and old individuals, which might have drove their tendency to be more generous in the three donation tasks. In terms of gender differences and age, women across all age groups donated more to charities than men across all age groups. Hence, age is one of the demographic factors that moderate the gender gap and differences in charitable giving. It was also found that income moderated giving behaviour and gender differences. Wealthier individuals were more generous in the donation compared to individuals earning less. Also, there was a linear relationship between income and donation amount, i.e., donation amount increases as a person earns more. In terms of gender differences and income, women were more generous than men across different income levels. This is consistent with past findings which show that women were more likely to plan, budget and donate for charitable causes. However, marital status and education did not have a significant effect giving behaviour. In this study couples and single individuals donated at the same level with no significant differences. These differences might have been moderated by other demographic characteristics in the study which were not controlled for when testing the difference between individual and couples giving behaviour. Therefore, contrary to prior findings, this study found that marriage was not positively associated with higher levels of giving. Lastly, there

was no significant interaction between education level and donation amount. Instead, it was found that donors who have completed a high school degree were more likely to donate at higher levels than donors who completed a bachelors or a master's degree. This contrasts with prior findings in research which have shown that more educated individuals are more likely to donate and donate higher amounts. Taken together, this study found that age and income moderated the difference and gender gap in giving behaviour, while education level and marital status did not have an effect.

Thirdly, the study investigated whether giving behaviour can be moderated by the type of charities shown to the donor. Specifically, donors in the study were shown gender-neutral charities that are multipurpose non-profit organizations that catered to receivers of all genders and gendered charities that focused specifically focus on female receivers. This is important because past research has shown that donors donate more to charities that cater to female receivers. However, contrary to prior research, this study found that both male and female donors donated more in the gender-neutral condition than the gendered condition. This phenomenon might be driven the fact that donors in the study didn't want to be egalitarian with their giving and spread their donation across multiple charities that focus on male and female receivers separately. Rather, donors gave to a small range of gender-neutral charities that cater to receivers of both genders together. This also suggests that both men and female donors were risk-seeking in their giving preferences as they focused on a single gender-neutral organizations rather than separate ones. Therefore, the results of this study show that donors donate more to gender neutral organizations than gender neutral organizations as they do not spread their donation thin across multiple organizations and rather focus on a single organization that caters to both genders.

Fourth, the effect that the number of contribution options have on prosocial choices was investigated. In line with prior research, this study found that the choice overload effect is reversed when donors make decisions that are beneficial for others. Donors in the study clearly preferred large choice sets over small choice sets as the mean level of giving was much higher in the former than the small choice set. Additionally, donors also reported higher levels of choice satisfaction in the large choice than the small choice sets which suggests that the probability of choice deferral was much less in the large choice sets. This happened because donors were promotion focused and perceived that large choice sets have more good options with positive outcomes as compared to small choices whereby positive outcome were absent because they had limited number of contribution options. Therefore, this study lends support for the promotion focused theory of choice overload which postulates that donors

feel more satisfied after donating in a large choice set as it gives them variety to choose from which has more positive outcomes associated with it.

Lastly, the study investigated whether information processing systems and gender differences in it measured by CRT have an influence on donation behaviour or not. A plethora of findings on information processing systems suggest that men tend to outperform women on the CRT which measures people's ability to engage in deliberate processing and dis-inhibit intuition across three numerical problems. Women tend to perform worse than men because they are more likely to experience mathematics anxiety which impairs performance on numerical performance. Contrary to prior findings, this study did not find a significant gender differences in CRT scores i.e., donors' ability to engage in deliberate reasoning. Additionally, donors CRT scores did not have a significant impact on giving behaviour, which suggests that information processing systems did not moderate giving behaviour among donors. This suggests that donors in the study evaluated all the contributions options across all the choice sets intuitively as a whole rather than deliberately assessing each option separately. Therefore, donors enacted prosocial behaviours intuitively that they had learned from previous similar experiences and internalizing social norms rather than engaging in a cognitively effortful process of analyzing each option separately to reduce the uncertainty related to the lack of information about the charities.

5.1 Theoretical Contribution

Taken together, these findings shed light on the gender gap in charitable giving and the effect of choice overload on giving behaviour. A plethora of research across psychology, economics and sociology has found that men tend to donate at lower levels than women because they are less empathetic. In the present, study it was found that men give less than women to charity, but this difference was not necessarily mediated by differences in empathy. On the contrary, this study is first to find to the best of our knowledge that lower empathetic men donate significantly more than lower empathetic women. Instead, this study found that demographic factors like age and income have a more robust effect in mediating differences in giving behaviour than empathy. This echoes prior research which have shown that age and income are significant factors that are implicated in giving behaviour. However, unlike previous research, this study did not find that more educated men and couples were more likely to donate. This contradictory finding suggests that more research with larger

representative samples need to be conducted to understand the effect of education , marital status and empathy on donation behaviour.

This study also investigated the key issue of giving to female-specific charities that has been largely understudied and only a few studies have investigated that because of the difficulty involved in separating these charities from survey responses. To examine this issue, this study used three choice task that varied in terms of the receiver it caters to. Specifically, donors either donated to gender neutral charities or gender specific charities. Using this new methodology, the study found that donation amount was significantly lower in the latter compared to the former. This finding has also contradicted past research which have shown that women donor specifically are more egalitarian in their giving and spread it across multiple charities. Therefore, these contradictory finding suggests that this question needs to be investigated in more detail using multiple methodology such as choice task, lab studies, field studies and surveys.

The study also contributed to the literature on choice overload in context of socially important goods. This study lends support for the promotion focused theory (Polman, 2012) which has shown that donors have a strong preference for large choice sets and more variety when choosing for others. Therefore, this study shows that promotion focused theory is a more robust predictor of giving behaviour than the choice overload theory which suggests that more options are detrimental to the person.

Lastly, the study also contributes to the literature on dual-process higher level cognitions. This study finds that men and women are not different in their ability to be analytical. This finding is consistent with a few studies in the past that have shown that men and women do not show differences in deliberative reasoning (Juanchich et al.,2016; Shiloh et al.,2002). Therefore, it is likely that the differences in CRT are driven by the superficial features of the testing element that triggers mathematics anxiety in women. Therefore, more research needs to be done to understand the gender gap in information processing system using novel methodologies (for a discussion on it, see Juanchich et al.,2019). Additionally, our study did not find a significant interaction between CRT scores and donation behaviour. This contradicts previous findings by Herzstein et al., (2020) , which show that information processing systems influence prosocial giving among donors. Therefore, this study provides contradictory findings which show that gender differences in information processing system found in the past might be moderated by the superficial elements of the CRT and there are no innate differences in men's and women's ability to act analytically. Additionally, more

research needs to be conducted in the future to understand the empirical linkage between information processing systems and donation behaviour.

5.2 Practical Contributions

Alongside theoretical contributions, this study also has clear practical implications that charities can consider when designing effective fundraising campaigns. Specifically, when designing campaigns, fundraisers should aim to provide more contribution options as donors seek for more variety when making these decisions. Also, fundraisers should avoid providing small and medium sets of contribution options because donors perceive that these choice sets have not been specifically curated by the charity which deters their motivation to donate at all. Lastly, fundraisers should aim to provide a lot of information about the charities, which will reduce the uncertainty about the options and help build more trust and social capital between the donor and receiver, resulting in higher number of donations.

5.3 Limitations

Despite the theoretical and practical contributions made by the study, there are several limitations associated with it. First and foremost, the study endowed donors with hypothetical rewards in the form of ECU's which do not capture the real motivation associated with planning, budgeting and donating real money. Secondly, the choice tasks used in the study relied on self-reported intentions about behaviours. While self-reported intentions are a robust predictor of behaviour (Fishbein and Ajzen, 2011), they are still a rough estimation of behaviour. Also, donors in the study might have experienced a social responsibility bias when completing the self-reported donation task. This social responsibility bias might have led donors to exaggerate their donation and giving behaviour to charities. Thirdly, the study did not make a clear distinction between the gender-neutral charities and gendered charities. Specifically, showing participants gender neutral charities in the gendered charities might have nudged them to contribute to those options rather than spread it across different charities. Fourthly, the study uses an arbitrary number of options across all the choice sets, leaving open the question on how many options are perceived to be large and small by the consumer and whether it varies as a function of the cultural context of the donor. Lastly, using the CRT to measure deliberative reasoning is problematic because past findings have

shown that it has low construct validity and it only measures a donors numeracy skill rather than actual differences in higher level cognitions (Zhang et al.,2016). Also in the study, the respondents completed the CRT after the donation task rather than before the task, which might be why the study did not find a significant effect of information processing system on donation behaviour.

5.4 Conclusion

To conclude, this study investigated the gender difference and effect of choice overload on donation behaviour using three different choice tasks. A gender gap exists in giving behaviour with women donating at slightly higher levels than men and the difference is moderated by demographic factors such as age and income. Additionally, donors' behaviour is not moderated by the type of receiver the charity is catering to. The study also found that donors experience higher choice satisfaction when choosing from large choice sets as opposed to small choices when making choices about others. This suggests that people become promotion focused when acting as a proxy decision maker for others. Lastly, the study found that information processing system does not mediate the gender gap or giving behaviour. Hence, donation behaviour was only moderated by demographic factors and the assortment size and further research is needed to understand what other factors are implicated in donation behaviour.

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7 Appendix A

Instrument

Appendix A describes the Cognitive Reflection Task in more detail. The CRT consists of three-numerical puzzles that require respondents to inhibit their intuition and engage more in deliberative reasoning whereby they have to calculate a sequence of computational processes to get the right results.

The following three numerical puzzles are used in CRT:

Q1- A bat and ball together cost £1.10. The bat costs one dollar more than the ball. How much does the ball cost?

Answer - 5 cents

Q2-f it takes 5 machines 5 minutes to make 5 widgets, how long would it take 100 machines to make 100 widgets?

Answer - 5 minutes

Q3-In a lake, there is a patch of lily pads. Every day, the patch doubles in size. If it takes 48 days for the patch to cover the entire lake, how long would it take for the patch to cover half of the lake?

Answer- 47 days

8 Appendix B

Instrument

Appendix B describes the Prosocial Personal Battery that can be used to measure people's ability to engage in empathy. It consists of 12 items that measure people's empathy, moral reasoning and social responsibility and self-reported altruism. The questions that were presented to the participants are shown below :

1- Empathy

Below are a number of statements that may or may not describe you, your feelings, or your behaviour. Please read each statement and pick the option that best fits you.

	Strongly Disagree	Disagree	Uncertain	Agree	Agree Strongly
I sometimes try to understand my friends better by imagining how things look from their perspective.	☐	☐	☐	☐	☐
I believe that there are two sides to every question and try to look at them both.	☐	☐	☐	☐	☐
When I'm upset at someone, I usually try to "put myself in their shoes" for a while.	☐	☐	☐	☐	☐

2-Social Responsibility

Below are a number of statements that may or may not describe you, your feelings, or your behaviour. Please read each statement pick the option that best fits you.

psv	Strongly Disagree	Disagree	Uncertain	Agree	Agree Strongly
I would feel less bothered about leaving litter in a dirty park than in a clean one.	☐	☐	☐	☐	☐
With the pressure for grades and the widespread cheating in school nowadays, the individual who cheats occasionally is not really as much at fault.	☐	☐	☐	☐	☐
When you have a job to do, it is impossible to look out for everybody's best interest.	☐	☐	☐	☐	☐

3-Moral Reasoning

Below are a set of statements, which may or may not describe how you make decisions when you have to choose between two courses of action or alternatives when there is no clear right way or wrong way to act. Please read each statement carefully and choose the option that best fits you.

	Does not describe me	Describes me slightly well	Describes me moderately well	Describes me very well	Describes me extremely well
I choose alternatives that are intended to meet everybody's needs.	☐	☐	☐	☐	☐
I choose a course of action that maximizes the help other people receive.	☐	☐	☐	☐	☐
I choose a course of action that considers the rights of all people involved.	☐	☐	☐	☐	☐

4-Self Reported Altruism

Below are a set of statements, which may or may not describe how you make decisions when you have to choose between two courses of action or alternatives when there is no clear right way or wrong way to act. Please read each statement carefully and choose the option that best fits you.

	Never	Once	More than once	Often	Very often
I have helped carry a stranger's belongings (e.g., books, parcels, etc.).	☐	☐	☐	☐	☐
I have allowed someone to go ahead of me in a line (e.g., supermarket, copying machine, etc.).	☐	☐	☐	☐	☐
I have let a neighbor whom I didn't know too well borrow an item of some value (e.g., tools, a dish, etc.).	☐	☐	☐	☐	☐

Appendix B also shows the questions that the study had used to measure choice satisfaction

Choice Satisfaction

Please read each statement carefully and choose the option that best fits you.

	Not at all	Extremely	In some cases
If you could put off the donation decision until a later date , how likely is it that you would do so?	☐	☐	☐
Did you find it difficult to make the donation decisions?	☐	☐	☐
Are you that you made the right choices in the tasks?	☐	☐	☐
Were making the choices frustrating for you?	☐	☐	☐